PROJECT PROPOSAL

BinGo

Smart Recycling & Waste Sorter

MODULE	Mobile Application Development
GROUP DETAILS	Group 120

TEAM MEMBERS

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PROBLEM DEFINITION

1. The Problem: Consumer Confusion & “Wish cycling”

The primary challenge in waste management is inaccuracy. Many consumers engage in “wish cycling” tossing non recyclable items into recycling bins hoping they are processed. This stems from complex packaging and obscure labeling, creating a high cognitive load that leads to incorrect disposal.

2. Consequences: Contamination & Error

User error at the point of disposal leads to recycling contamination. Items like greasy pizza boxes or incompatible plastics can ruin entire batches.

- **Operational Failure:** Contaminated batches are rejected and sent to landfills, nullifying recycling efforts.
- **Economic Loss:** Manual sorting and disposal of contaminated waste impose significant financial burdens on municipalities.

3. The Localization Gap

Recycling standards are hyper local; what is recyclable in one city may not be in another. Traditional solutions (posters or generic searches) lack the real time, location aware support needed for accurate disposal decisions.

4. Project Significance & Technology

Solving these issues is vital for a Circular Economy:

- **Environmental Impact:** Reducing contamination increases material yield and lowers carbon emissions.
- **Behavioral Change:** Leverages Computer Vision and Geolocation to provide “just in time” learning, moving beyond passive education to active habit formation.
- **Modern Relevance:** High-accuracy object detection and GPS services now allow for the hyper local, real time guidance previously impossible.

PROJECT SCOPE STATEMENT

1. Project Objectives

BinGo is a cross platform mobile app leveraging AI to help users identify and sort household waste correctly.

- **Accuracy:** High confidence classification of common waste materials via image recognition.
- **Localization:** Location specific disposal instructions based on municipal rules.
- **Engagement:** Incentivized recycling via points and leaderboards.

2. In Scope Features

- **BinGo Vision:** Real time AI analysis and classification (Recyclable, Compost, E-Waste, Landfill) with preparation warnings.
- **Geolocation:** GPS based rule adaptation and an interactive map of nearby recycling centers.
- **Gamification:** User profiles, “BinGo Points” for scans, and local community leaderboards.
- **Education:** A searchable database for manual lookup and a feed of eco tips.

3. Out of Scope (Exclusions)

- **Industrial Waste:** Limited to household consumer waste.
- **Hardware:** Software only solution; no smart bins or physical sensors.
- **Live Logistics:** No integration with truck schedules or on-demand pickup services.
- **E Commerce:** No marketplace for recycled goods.

4. Deliverables

- Functional mobile application branded “BinGo”.
- AI model for 20+ waste categories.
- Backend database for users, centers, and rules.
- Technical documentation, API guides, and user manuals.

5. Assumptions & Constraints

- **Assumptions:** Stable internet for AI processing; availability of third party Maps/AI APIs.
- **Constraints:** Dependent on public municipal data; recognition may struggle with dirty or obscure items.

FEATURES OF THE APPLICATION

1. BinGo Vision (AI Scanner)

The AI powered camera removes guesswork from recycling with real time detection and material analysis.

- **Instant Sorting:** Color coded cards (Green: Compost, Blue: Recyclable, Black: Landfill, Red: Special) provide immediate guidance.
- **Preparation Steps:** Displays localized “Prep Steps” (e.g., “Rinse container,” “Remove cap”) to ensure batch purity.

2. Location Based Intelligence

BinGo adapts to varying municipal rules using GPS auto-switching.

- **Interactive Eco Map:** Locates public recycling bins, E waste centers, donation points, and hazardous waste sites.
- **Facility Details:** Provides operating hours, accepted materials, and turn by turn navigation.

3. Gamification & Engagement

Recycling is transformed into a rewarding experience.

- **Points & Challenges:** Earn “BinGo Points” for scans and complete daily/weekly missions.
- **Community:** View local leaderboards and unlock “Impact Badges” (e.g., “Glass Guardian”) for consistent recycling milestones.

4. Educational Hub (Waste Wiki)

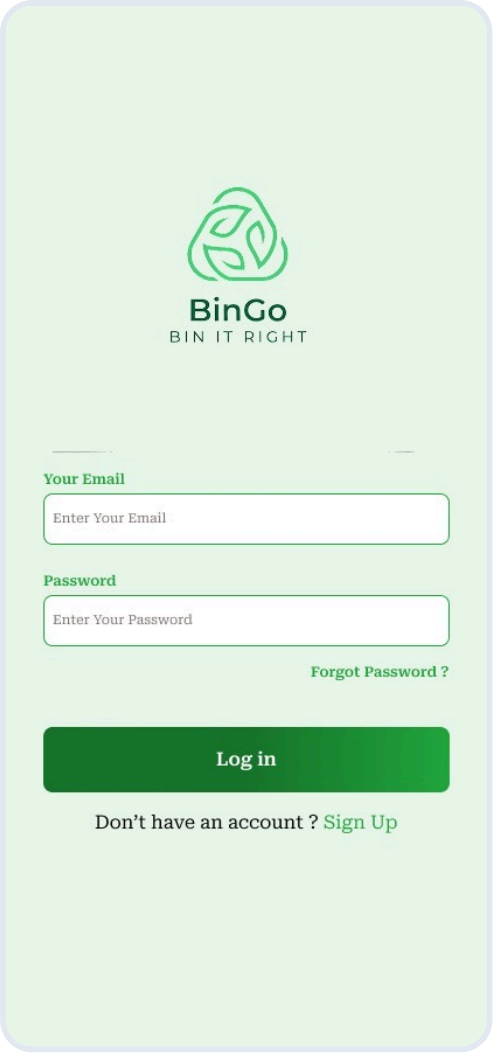
- **Searchable Database:** Manual lookup for complex items (e.g., “How to recycle a mattress?”).
- **Daily “Did You Know?”:** Tips and facts to bust common recycling myths and reduce waste at the source.

5. User Profile & Impact

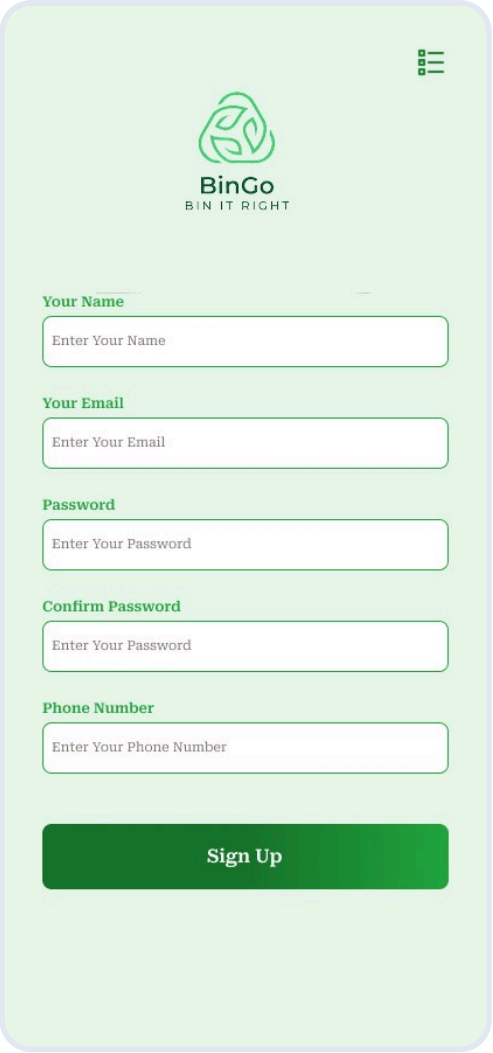
- **Impact Dashboard:** Visualizes environmental contributions (e.g., “Trees saved,” “Plastic diverted”) and tracks full scan history.
- **Leader Dashboard:** A central hub for live global/local rankings, rewards redemption, and group challenge tracking.

MOBILE APP WIREFRAMES

1. Authentication & Onboarding

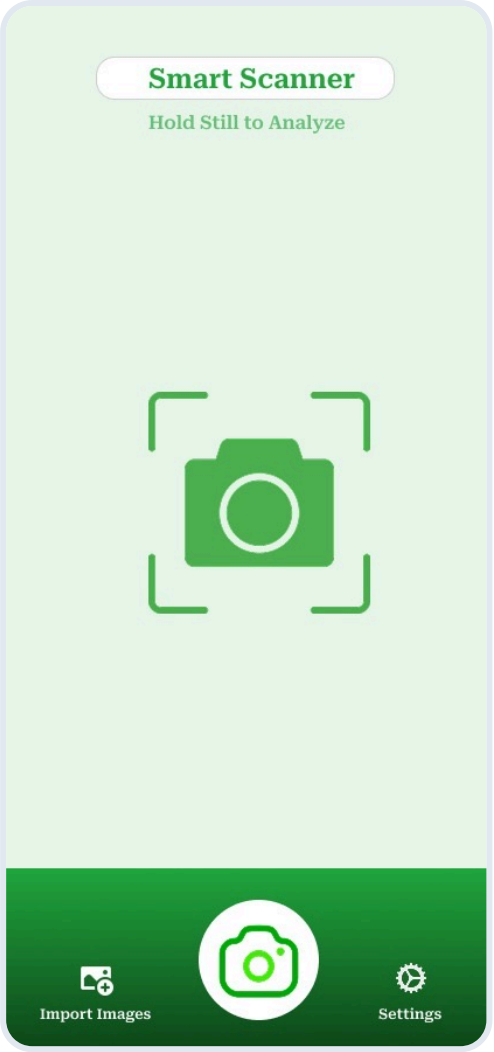


LOGIN SCREEN

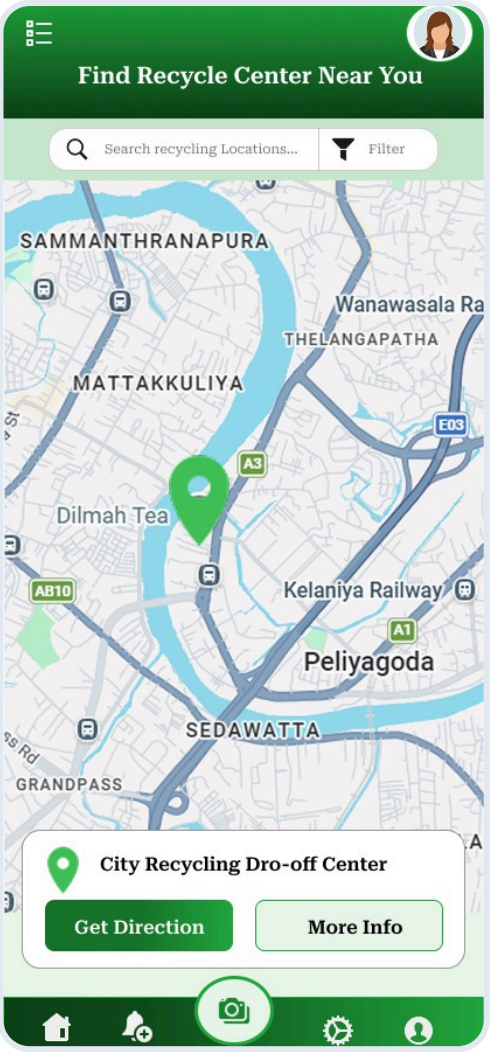


SIGN UP SCREEN

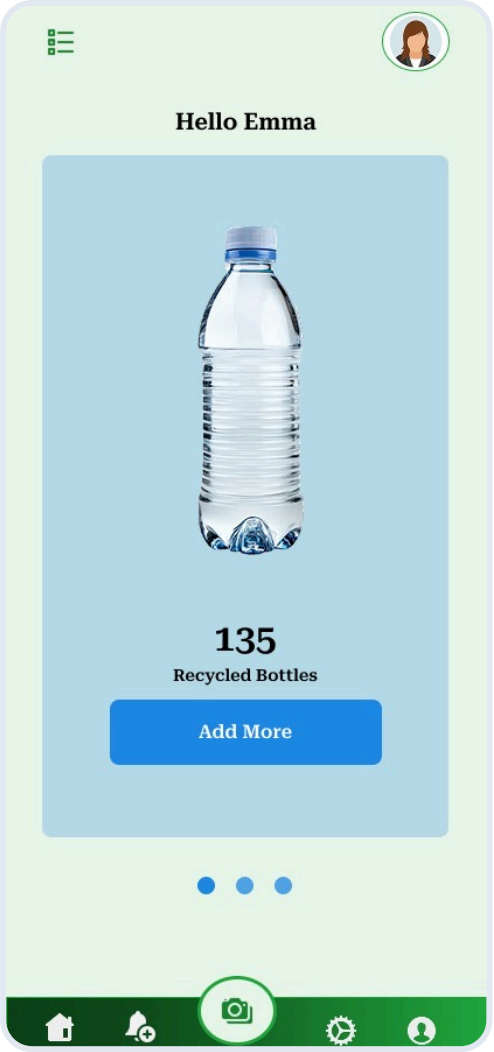
2. Core Navigation & Features



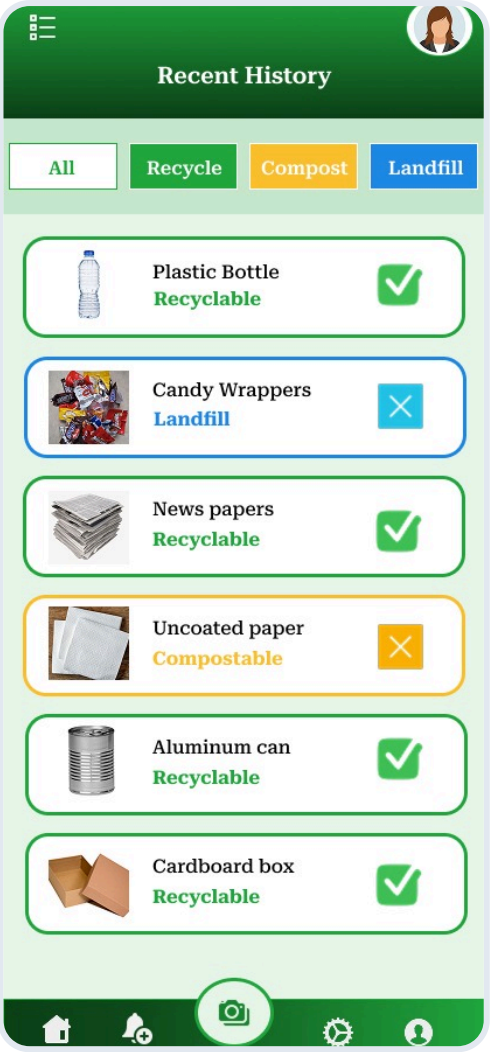
HOME DASHBOARD



RECYCLING CENTER MAP

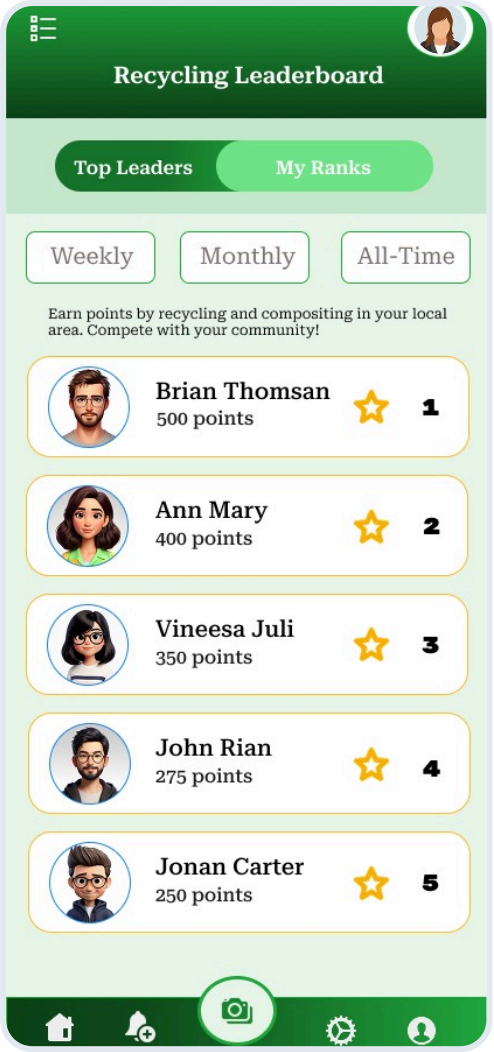


ITEM SCANNING & IDENTIFICATION



USER SCANNING HISTORY

3. Engagement & Profile



COMMUNITY LEADERBOARD

USER PROFILE & STATS

Thank You!